PRIMARY TEACHING NOTES SERIES

Teaching Mathematics in Primary Classes

How to build number sense, teach the four operations step by step, and make maths friendly for young children - practical guidance for primary teachers.



By **Shah Fahad**
Primary School Teacher (PST), ESED Khyber Pakhtunkhwa - Buner, Pakistan
November 2026

These are study and practice notes for primary teachers, written in simple English. They cover one topic only - teaching mathematics in primary classes - and follow the themes taught in teacher induction training: how young children learn maths, number sense, place value, the four operations, fractions, shapes, measurement, time, money, word problems, and checking learning without fear.

> **Plain idea:** In primary classes, maths is not learned by watching the teacher solve sums on the board. It is learned by touching, counting, grouping, drawing and talking - first with real things, then with pictures, and only then with written symbols.

Why Mathematics Matters - and How Children Learn It

Mathematics is the subject children use every day without noticing: counting money, sharing food fairly, telling the time, measuring cloth, checking change at a shop. A child who is confident in primary maths carries that confidence into science, work and daily life. A child who falls behind early often stays behind, because every new topic stands on the old one.

The three steps: concrete, pictorial, abstract

Induction training gives one rule above all others for primary maths: **concrete - pictorial - abstract**, in that order, every time a new idea is taught.

1. **Concrete - real things.** Children touch and move real objects: stones, sticks, bottle caps, pencils, blocks. "Show me 7" means seven real things on the desk.
2. **Pictorial - pictures.** Children draw or look at pictures of the things: circles, tallies, dots, a number line. The object is gone, but its picture remains.
3. **Abstract - symbols.** Only now come the written signs: 7, +, -, =. The symbol now stands for something the child has already touched and drawn, so it has meaning.

> **Common mistake:** Starting at step 3. When a Class 1 child is given "5 + 3 = ?" before ever joining two real groups of objects, the sum is only marks on paper. The child memorises, forgets, and begins to fear maths. Go back to real objects - the fear usually goes with it.

What good primary maths teaching looks like

- **Children do, teacher guides.** In a good period, children are counting, grouping, drawing and explaining more than the teacher is talking.
- **Talk is part of maths.** Ask "How did you get that answer?" A child who can explain a method understands it; a child who only repeats it may not.
- **One idea, fully, before the next.** Maths is a staircase. Skipping a weak step makes every later step shaky.
- **Mistakes are information.** A wrong answer shows the teacher exactly which step to re-teach. Never punish a maths mistake - use it.
- **Daily life everywhere.** Use prices, bus times, cricket scores, dates and classroom objects so children see maths as useful, not distant.

Number Sense and Counting

Number sense means a child truly understands what a number is - how big it is, what comes before and after it, and how it can be split and joined. It is the foundation of all later maths, and it is built in Classes 1 to 3 through counting with understanding, not counting by rote alone.

Counting that means something

- **One-to-one:** the child touches or moves one object for each number word - one stone, "one"; next stone, "two".
- **Stable order:** the counting words always come in the same order - 1, 2, 3, 4, 5.
- **The last word is the total:** after counting 8 pencils, "eight" answers "How many pencils?" Many young children recount instead of answering - teach this point directly.
- **Any order works:** objects spread out, in a line, or in a circle still give the same total. Let children test this themselves.
- **Counting on and back:** from any number, not only from 1. "Start at 6 and count 4 more" is a key early skill for addition.

Activities that build number sense

- **Counting collections:** each child counts a small pile of stones or caps, then a partner checks. Change the pile size to match the child's level.
- **Number line on the floor:** chalk a line 0 to 20 in the yard. Children walk it - forwards, backwards, and jumping in twos.
- **Before, after, between:** quick oral questions all day - "What comes after 17? Before 30? Between 44 and 46?"
- **Comparing numbers:** two piles, two numbers - which is more, which is less, and by how much? Introduce the words *greater than*, *less than* and *equal to* with real piles first.
- **Odd and even:** pair up the objects. If one is left alone, the number is odd. Children discover the pattern themselves.
- **Skip counting:** in twos, fives and tens, clapping or stepping as they count - this quietly prepares multiplication tables.

> **Teacher's check:** Give a child 13 counters and ask three questions - "How many are there? Give me 9. Put them in two groups and tell me the groups." The answers show in one minute whether counting, number recognition and splitting numbers are secure.

Place Value, Addition and Subtraction

Place value is the backbone of written maths: the same digit 4 means four in 4, forty in 47, and four hundred in 452. Children who do not understand this write sums correctly by luck and wrongly by habit. Build it with bundles before you build it with columns.

Teaching tens and ones

- Use sticks or straws. Ten loose sticks tied with a rubber band make **one ten**. Children physically make 34 as 3 bundles and 4 loose sticks.
- Show the exchange both ways: 10 ones = 1 ten, and 1 ten opened = 10 ones. This single idea explains both carrying and borrowing later.
- Use a simple place-value chart on the board - Hundreds | Tens | Ones - and always place the bundles under it before writing digits under it.
- Say numbers in expanded form: 47 is "40 and 7". Children who can say this rarely write 47 as 407.

Addition, step by step

1. Join two real groups and count all - then count on from the bigger group.
2. Draw the groups (dots or tallies) and count the drawing.
3. Write the vertical sum, ones under ones, tens under tens.
4. Carrying: when the ones make ten or more, bundle the ten and carry it to the tens side - show the bundle moving, do not just say "carry the 1".

Subtraction, step by step

1. Take away real objects and count what remains.
2. Count back on the number line - subtraction as moving backwards.
3. Write the vertical sum with ones under ones.
4. Borrowing: when the ones are not enough, open one ten-bundle into 10 ones. Children who have opened real bundles accept this at once; children who have only heard "borrow 1" often write the smaller digit on top.

> **Watch for:** reversed digits (writing 12 as 21), adding tens to ones, and subtracting the smaller digit from the bigger in every column. Each error points back to place value - return to bundles, not to more drill.

Multiplication, Division and Tables

Multiplication and division are two views of the same idea. Teach them together, from groups of real objects, and tables will grow out of understanding instead of sitting on top of fear.

Multiplication is groups

- Start with equal groups: 4 plates with 3 stones each. Children count all the stones, then count in threes: 3, 6, 9, 12.
- Draw the groups as an array - rows and columns of dots. 3 rows of 4 and 4 rows of 3 give the same total; children see why 3 x 4 = 4 x 3.
- Only then write the sign: 4 x 3 = 12, read as "4 groups of 3 make 12".
- Connect it to repeated addition: 3 + 3 + 3 + 3 = 12. Both forms stay on the board together for weeks.

Division is sharing and grouping

- **Sharing:** 12 sweets shared fairly among 4 children - deal them out one by one, like cards. Each child gets 3.
- **Grouping:** 12 sweets, put 3 in each packet - how many packets? Count the groups: 4.
- Show the family together: 4 x 3 = 12, so 12 divided by 4 = 3 and 12 divided by 3 = 4. A child who knows one fact owns three.
- Remainders come naturally from sharing: 13 sweets among 4 children leaves 1 left over. Name it only after children have seen it happen.

Learning tables without fear

- Build each table from skip counting first - in twos, fives, tens, then threes and fours.
- Show patterns: the 5 table ends in 5, 0, 5, 0; the 9 table's digits add to 9. Patterns are remembered when lists are forgotten.
- Practise little and often: two minutes of oral tables daily beats one long weekly drill.
- Use quick games - table bingo, "beat your own score", partner flashcards - and praise speed gains, never shame slow children in front of the class.

Fractions, Shapes and Measurement

Fractions begin with fair sharing

A fraction is a fair share of a whole. Begin with a roti, an apple or a paper circle shared between two children, then four. The language comes first - half, quarter, one third - and the written form 1/2 comes after the child has folded, cut and compared real pieces many times.

- Fold paper circles and strips into 2, then 4, then 8 equal parts. Unequal parts are *not* fractions - let children reject unfair pieces themselves.
- Compare with pieces in hand: one half is bigger than one quarter, because fewer sharers means bigger shares. This surprises children and fixes the common error "4 is bigger, so 1/4 is bigger".
- Use fractions of groups too: half of 10 pencils, a quarter of 12 stones.
- Keep it practical in Classes 1 to 3: halves, quarters and thirds with objects and drawings; formal fraction sums belong to later classes.

Shapes children can touch

- Teach 2D shapes (circle, triangle, square, rectangle) from classroom objects - a clock, a book, a set square - before shape names in a book.
- Count sides and corners together by running a finger along each edge.
- Introduce 3D shapes (cube, cuboid, sphere, cylinder) through boxes, tins and balls, and let children sort objects by shape in a game.
- Draw, trace and make shapes with sticks and thread. A shape a child has built is a shape the child remembers.

Measurement: compare first, units second

- **Length:** first compare (longer, shorter), then measure with hand spans and sticks, then meet the ruler, metre and centimetre. Children who start with non-standard units understand why a standard unit is needed.
- **Weight:** heft two objects, guess, then check on a balance - heavier, lighter, about the same; kilograms come later.
- **Capacity:** fill bottles and jugs with cups of water at a tap or bucket. "How many cups fill this bottle?" is real maths.
- Always estimate first, measure second. Estimation is where number sense and measurement meet.

Time, Money and Simple Data

Telling the time

- Start with the day itself: morning, noon, evening, night, and the order of the school day. Then days of the week, months of the year, and the calendar.
- Teach the clock in steps: o'clock first (long hand on 12), then half past, and only later quarter past and quarter to.
- Make a large cardboard clock with moving hands. Every child must set times and read times on it - reading a printed clock picture is not enough.
- Ask time questions all day: "Break starts at 10:30. What will the clock look like?"

Money is daily maths

- Use real or play coins and notes: 1, 2, 5, 10 rupees and larger notes as classes progress. Children sort them, name them, and make a given amount in different ways - 20 rupees as two 10s, or four 5s.
- Run a classroom shop: priced pencils, erasers and sweets (on paper). Buying, paying and checking change teaches addition and subtraction with real purpose.
- Teach change as counting up: from Rs 37 to Rs 50 is 3 more to 40, then 10 more - Rs 13 change. This is how shopkeepers actually do it, and it is easier than borrowing.

Handling simple data

- Collect class data children care about: favourite fruit, how each child travels to school, number of siblings.
- Record it first with real objects or pictures in rows - a **pictograph** - then as a simple bar chart with one square per child.
- Read the chart together: Which is most? Which is least? How many more? How many altogether?
- Keep one class chart on the wall and change its question every month. Data handling is a habit, not a chapter.

> **Linking the topics:** A classroom shop uses money, addition, subtraction and change in one activity. A class pictograph uses counting, comparing and totals. Plan activities that join topics - real life never separates them.

Word Problems, Mental Maths and Common Mistakes

A routine for word problems

Many children can do a sum but freeze before a word problem - the difficulty is reading, not maths. Teach one routine and use it every time:

1. **Read** the problem aloud, slowly, twice.
2. **Find** what is given and what is asked. Children underline the numbers and circle the question.
3. **Draw** it - objects, boxes or a line. The drawing shows whether the story means join, take away, groups, or share.
4. **Choose** the operation and say why in one sentence.
5. **Solve**, then **check**: does the answer fit the story? 5 mangoes plus 3 cannot make 2.

Use stories from the children's own world - fruit from the bazaar, runs in cricket, passengers in a bus. Change only the numbers in a familiar story until the routine is secure, then change the story.

Mental maths: five minutes daily

- Number bonds to 10 and 20 (7 + 3, 13 + 7) until they are instant - they carry all later written methods. Add and subtract tens and hundreds orally too: 40 + 30, 90 - 20, 300 + 400.
- Doubling and halving: double 15, half of 30 - quick, useful, and it prepares fractions and tables. Keep it oral and fast, with slates for answers, so every child answers every question.

Common mistakes and what they mean

What you see	Likely cause	Fix
Subtracts the smaller digit from the bigger in each column	Borrowing not understood	Return to opening ten-bundles; exchange before writing
Writes 47 as 407, or reads 305 as 35	Weak place value	Expanded form aloud; place-value chart for every sum
Chooses the wrong operation in word problems	Guessing from keywords	Draw the story first; choose the operation from the drawing
Counts on fingers for every small sum in Class 4-5	Number bonds not secure	Daily number-bond practice in pairs, short and regular
Right in class, wrong in tests	Speed pressure, fear	Untimed practice first; timed only when accurate

Maths Fear, Assessment and a Weekly Routine

Removing the fear of maths

Many children decide by Class 3 that they are "not a maths person". That decision is taught, not born - by rushed symbols, public shaming, and sums above the child's step. Reverse it:

- Praise the method and the effort, not only the right answer: "Your drawing showed exactly what to do."
- Never read out marks or ranks in maths. Compare a child only with their own last week.
- Let children answer on slates held up together, so a wrong answer is private between child and teacher.
- Give every child sums at their staircase step. Success at the right step builds courage for the next one - and show that the teacher also checks and corrects. Maths is checking, not magic.

Checking learning, little and often

- **Daily:** oral questions, slate work and a walk around the class while children work - catch the error in the period, not in the test.
- **Weekly:** a short written check of the week's one or two ideas only, marked the same day, with the weakest idea re-taught next week.
- **Per term:** note which children are secure, shaky or lost on each big idea - number, place value, each operation, fractions, measurement - and plan groups from that note. Assessment's job is to tell the teacher what to teach next.

A simple weekly rhythm

Day	Focus
Day 1	New idea with real objects (concrete step) + mental maths warm-up
Day 2	Pictures and drawings of the same idea (pictorial step)
Day 3	Written sums and textbook practice (abstract step)
Day 4	Word problems and a practical activity - shop, measuring, or chart
Day 5	Games, tables practice, weekly check, re-teach the weakest point

Teacher's closing checklist

- Did the new idea start with real objects before symbols?
- Did every child handle, draw or say something in the period - not only watch?
- Did I ask "how did you get it?" at least three times?
- Do I know which children are lost on this idea, by name, and is tomorrow's first task set at their step?